

# Ansh Tyagi

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## EDUCATION

### Stony Brook University

Expected Graduation: May 2027

*Master of Science in Computer Science (Advisor: Prof. Erez Zadok), GPA: 4.0/4.0*

*Stony Brook, NY*

- Relevant Coursework: Distributed Systems, Database Systems, Vision Language Models, Data Science Fundamentals

### Indian Institute of Information Technology, Sonapat

August 2019 – May 2023

*Bachelor of Technology in Information Technology, CGPA: 9.35/10*

*Sonapat, India*

- Relevant Coursework: Data Structures & Algorithms, Operating Systems, Computer Networks, DBMS, Machine Learning

## TECHNICAL SKILLS

**Programming Languages:** Kotlin, Java, Rust, Go, Python, SQL, C/C++, JavaScript, Dart

**Databases:** PostgreSQL, MariaDB, MongoDB, Redis, DynamoDB

**Frameworks & Libraries:** Spring Boot, Kafka, RabbitMQ, TensorFlow, OpenCV

**APIs & Protocols:** REST, GraphQL, gRPC, OAuth 2.0, OpenID Connect (OIDC)

**Cloud & DevOps:** Amazon Web Services (AWS), Google Cloud Platform (GCP), Kubernetes, Docker, Prometheus, Grafana, Jenkins

**ML & Inference:** LLMs, Vision-Language Models, KVCache optimization, Inference optimization, GPU Infrastructure, Model benchmarking

**Architectures & Practices:** Microservices, Event-Driven, Distributed Systems, Test-Driven Development, CI/CD, Observability

## WORK EXPERIENCE

### Intuit

May 2026 – August 2026

*Software Engineer Intern – Identity & Session Platform Team*

*Mountain View, CA, USA*

- Engineer services on Intuit's Identity & Session Platform powering **SSO** across TurboTax, QuickBooks, Credit Karma, and Mailchimp, spanning billions of token-validation requests for **100M+ customers** at **99.9%+ availability**.
- Reduced **p99** latency from **4.2s** to **300ms** for **bulk authorization** in **Intuit Enterprise Suite (IES)**, scaling to **5,000 accounts/user** at **200 TPS** via a rate-limited async batched-write path that removed **DynamoDB** hot-partition throttling.
- Eliminated deployment-related 503 errors (**60+ to 0** per rollout) by adding graceful shutdown via **Kubernetes** preStop hooks.

### Headout

January 2022 – July 2025

*Senior Software Engineer (April 2025 – July 2025)*

*Bengaluru, India*

- Architected Headout's Creators Platform backend from scratch as sole engineer (**Spring Boot, Kafka, Redis**), cutting onboarding from **13 days to 10 minutes** and driving **collaboration growth from 50 to 700 creators per quarter**
- Reduced creator impression cost **80%** and raised 15-day content completion to **50%** by owning the tool's event-driven content pipeline and economics.
- Built Offers Management for supply partners on an inventory-and-pricing data model with configurable discount controls, growing conversion up to **30%** and reaching **66% adoption** within a month while reducing discount-related support tickets **12%**.

*Software Engineer (May 2023 – April 2025)*

- Slashed listing time **95% (72 to 4 minutes)** by building Magic Listing, an **AI driven 3-step workflow** replacing 60+ manual fields via **adapter-pattern** layer automating third-party vendor **API** ingestion and field mapping across the catalog.
- Trimmed schedule-request turnaround **40%** and communication errors **60%** with a Novu-based notification system.
- Lowered processing time **33%** and manual effort **25%** (90 requests/month) by shipping a self-serve Schedule Management & Moderation Layer with one-click supplier-to-Hub migration.

*Software Engineer Intern (January 2022 – May 2023)*

- Achieved **96% code / 87% branch coverage** by enforcing **TDD** as an early backend engineer on **Hub**, Headout's B2B platform on **Spring Boot WebFlux** with **REST & GraphQL** APIs.
- Boosted order success rate **80%** by re-architecting the London Theatre Direct integration plugin as its development owner.
- Improved operational efficiency **30%** by spearheading Native Listing, enabling direct supply-partner integration.

## PROJECTS & OPEN SOURCE

### Column-Delta Multi-Version Concurrency Control | *Distributed Database, Mako, C++* | [GitHub](#)

- Devised a schema-aware MVCC optimization in Mako that stores only modified columns, contributing **481 lines** of reusable **C++** that cut MVCC storage by **69%** on TPC-C while staying within **1%** of baseline throughput.

### VLFit | *Python* | [GitHub](#)

- Curated a vision-language benchmark for fit-in spatial reasoning with **220 image-question pairs** across 60–70 categories; of **6 state-of-the-art VLMs**, the best reached only **60.9%** and numeric reasoning fell to **5.7%–22.9%**.

### fxlang | *Rust* | [GitHub](#)

- Built an interpreted object-oriented language from scratch in **2,300 lines of Rust**, spanning the full pipeline from lexing and parsing through static resolution to a tree-walking interpreter (classes, inheritance, closures, lexical scoping).

### Open Source Contributions | *Kotlin & Rust ecosystems* | [GitHub](#)

- Contributed **16 merged PRs** – **gccrs** (GCC Rust frontend, **2.9k stars**): type system and CI; **KotlinDL** (**1.6k stars**): 9 PRs adding neural-network layers; **Multik** (**730 stars**), **KMath** (**720 stars**): ndarray and fast-math utilities; **nixpkgs** (**25k stars**): packaged zellij.

## LEADERSHIP, AWARDS & PUBLICATIONS

- Teaching Assistant**, Stony Brook University – CSE 416 (Software Engineering) and ISE/POL 369; code reviews, recitations, grading, mentoring *January 2026 – May 2026*
- Award: Smart India Hackathon 2022 Winner** – National-level hackathon organized by the Government of India *2022*
- Publication:** Tyagi, A., et al. “Design and Implementation of Dynamic Lightweight Elliptic Curve Cryptography,” COMS2, Springer *February 2021*